

Industrial Data Dashboard for Predictive Maintenance with Intelligent Recommendation System

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Abstract— Industrial systems require continuous monitoring and intelligent decision-making to ensure operational efficiency and equipment reliability. This paper presents an AI-integrated Industrial Data Dashboard and Predictive Maintenance System enhanced with a GPT-based Intelligent Recommendation Engine. The proposed framework enables real-time acquisition of electrical and machine condition parameters, including voltage, current, power, energy consumption, temperature, and vibration, using industrial sensors and energy meters. The collected data is processed and stored in a centralized database and visualized through an interactive Power BI dashboard for comprehensive trend analysis and performance monitoring.

A predictive maintenance model is implemented to compute degradation indices and normalized risk scores based on machine condition parameters, enabling early detection of abnormal operating conditions. Furthermore, a GPT-based recommendation module analyses summarized dashboard metrics and generates contextual suggestions for energy optimization, maintenance scheduling, and operational improvements. Experimental validation demonstrates reliable real-time monitoring, accurate anomaly detection, and enhanced decision support capabilities. The proposed system provides a scalable, cost-effective, and Industry 4.0-aligned solution that integrates IoT, business intelligence, predictive analytics, and generative AI for intelligent industrial management.

Keywords— AI-Integrated Industrial Monitoring, Predictive Maintenance Analytics, GPT-Based Recommendation System, Intelligent Energy Optimization, Power BI Industrial Dashboard, Machine Health Indexing, Industry 4.0 Decision Support, Smart Condition Monitoring Framework.

I. INTRODUCTION

Energy efficiency has emerged as a critical objective in modern industrial operations due to rising energy costs, increasing global demand, and stringent environmental regulations. Industrial sectors account for a substantial share of global electricity consumption, and inefficient energy utilization directly affects operational expenditure, system reliability, and environmental sustainability. Consequently, industries require intelligent monitoring frameworks capable of providing continuous visibility into energy consumption patterns and system performance.

Conventional energy monitoring systems rely primarily on periodic manual readings and static reporting mechanisms. Such approaches offer limited real-time insight and are inadequate for detecting peak demand conditions, abnormal consumption behavior, and operational inefficiencies. The absence of integrated analytical tools further restricts the ability of decision-makers to interpret large volumes of industrial data effectively.

Recent advancements in Industrial Internet of Things (IIoT), cloud computing, and data analytics have enabled the development of smart monitoring architectures. Industrial sensors and energy meters can continuously acquire electrical parameters such as voltage, current, active power, and energy consumption. These parameters can be transmitted to centralized storage systems for processing and analysis. Visualization platforms, including Power BI, facilitate the transformation of raw sensor data into meaningful insights through key performance indicators (KPIs), trend analysis, and interactive dashboards.

This paper proposes an AI-integrated Industrial Data Dashboard and Smart Energy Monitoring System designed to enhance energy efficiency and operational transparency. The proposed framework integrates real-time data acquisition,

structured data processing, and interactive visualization to support informed industrial decision-making. The system is scalable, cost-effective, and aligned with Industry 4.0 principles for intelligent industrial management.

II. LITERATURE REVIEW

The development of intelligent industrial monitoring systems has been strongly influenced by advancements in the Internet of Things (IoT) and data-driven architectures. Zanella *et al.* [1] presented foundational concepts of IoT-based smart environments, emphasizing real-time data acquisition, interoperability, and scalable communication frameworks. These principles form the basis for modern industrial monitoring infrastructures.

Karnouskos [2] highlighted the importance of Industrial IoT (IIoT) in energy management systems, demonstrating how distributed sensing and cyber-physical integration enable optimized industrial energy utilization. The study emphasized the role of real-time data analytics in improving operational efficiency and reducing energy wastage. Buyya and Dastjerdi [3] further discussed IoT principles and cloud-based processing paradigms, supporting centralized data storage and dashboard-driven industrial analytics.

Dashboard-based energy monitoring systems have been explored to enhance industrial transparency. Ahmed *et al.* [4] proposed an IoT-based real-time energy monitoring framework integrated with visualization dashboards, showing that the presentation of key performance indicators (KPIs) significantly improves operational awareness. Zhou *et al.* [5] investigated machine learning approaches for energy consumption prediction and anomaly detection in smart industries, demonstrating the effectiveness of data-driven models for intelligent decision-making. Additionally, the International Energy Agency (IEA) report [6] emphasizes the critical role of digital monitoring systems in achieving industrial energy efficiency and sustainability targets.

Advanced analytics and intelligent modelling techniques have been explored in various engineering domains. Pashikanti [7]–[10] demonstrated the application of deep learning, optimization algorithms, and adaptive analytical frameworks for complex signal analysis and classification tasks. Although these studies focus on biomedical signal processing, the underlying methodologies—such as feature extraction, optimization-based modeling, and neural network-driven classification—provide valuable insights into the development of predictive and intelligent analytical systems. These techniques highlight the potential of integrating advanced computational intelligence within industrial monitoring frameworks.

While prior research has addressed IoT-based monitoring, predictive analytics, and dashboard visualization

independently, limited work has explored the integration of industrial dashboards with AI-driven recommendation engines. In particular, the combination of real-time monitoring, predictive maintenance modeling, and generative AI-based advisory systems remains underexplored. The proposed study aims to bridge this gap by integrating IIoT-enabled energy monitoring, Power BI-based visualization, predictive analytics, and a GPT-based intelligent recommendation engine into a unified and scalable industrial framework.

III. PROBLEM STATEMENT

Industrial plants consume substantial amounts of electrical energy, and inefficient utilization leads to increased operational costs, accelerated equipment degradation, and negative environmental impact. In many industries, energy monitoring systems remain fragmented and partially manual, limiting the ability to obtain continuous and comprehensive insights into energy usage patterns.

Traditional monitoring approaches rely on periodic data collection and static reporting, which are insufficient for identifying peak demand conditions, abnormal consumption behavior, and emerging system inefficiencies. The absence of real-time analytics restricts proactive decision-making and often results in delayed corrective actions.

Furthermore, the lack of integrated data dashboards and intelligent analytical frameworks makes it challenging for operators and management to interpret large volumes of industrial data effectively. Without automated interpretation and contextual recommendations, decision-making remains dependent on manual analysis and domain expertise.

Therefore, there is a need for a unified, scalable, and intelligent monitoring system that integrates real-time data acquisition, interactive dashboard visualization, predictive analytics, and AI-driven recommendation capabilities to support efficient, sustainable, and data-driven industrial energy management.

IV. METHODOLOGY

The proposed system architecture is designed as a multi-layered framework integrating data acquisition, communication, storage, analytics, visualization, and AI-based recommendation components. The architecture ensures seamless data flow from industrial sensors to decision-support outputs while maintaining scalability and real-time responsiveness.

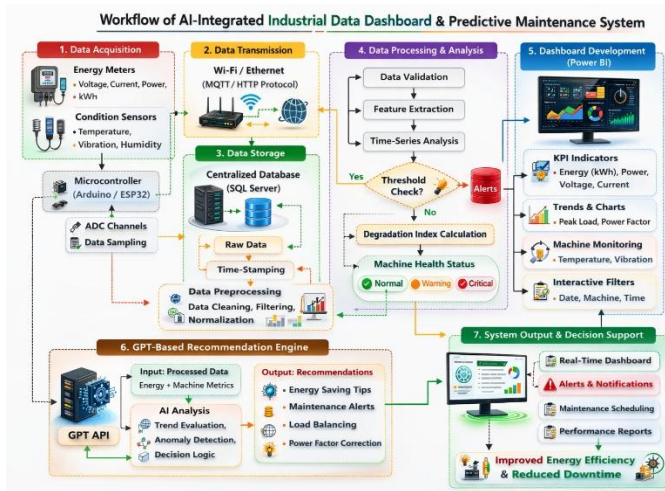


Fig.1 System Architecture

The methodology adopted for the Industrial Data Dashboard and Smart Energy Monitoring System is designed to ensure accurate data acquisition, reliable processing, effective analysis, and intuitive visualization of industrial energy consumption. The overall workflow is divided into multiple stages, from sensor-level data collection to dashboard-based decision support.

A. System Initialization and Configuration

At the initial stage, the industrial setup is analysed to identify critical energy consumption points such as machines, production lines, and utility systems. Smart energy meters and industrial sensors are configured with appropriate measurement ranges, sampling intervals, and communication parameters. Threshold limits for voltage, current, and power consumption are defined based on industrial standards and operational requirements.

B. Data Acquisition and Sampling

Electrical parameters including voltage, current, active power, reactive power, power factor, and energy consumption are measured continuously using calibrated sensors and smart meters. Data is sampled at fixed time intervals to capture both steady-state and transient energy behaviour. Time-stamping is applied to each data sample to support accurate temporal analysis and synchronization across multiple data sources.

C. Data Transmission and Communication

The collected sensor data is transmitted to a centralized server or cloud platform using secure and reliable communication protocols such as Ethernet, Wi-Fi, or industrial communication standards. Data transmission mechanisms are designed to minimize latency and packet loss, ensuring near real-time availability of energy data for analysis and visualization.

D. Data Storage and Preprocessing

Incoming data is stored in a structured database optimized for time-series data. Preprocessing techniques such as data validation, noise filtering, normalization, and missing value

handling are applied to improve data quality. Redundant or inconsistent records are removed to ensure reliable analysis and accurate visualization.

E. Data Analytics and Energy Assessment

The processed data is analysed to derive meaningful energy metrics, including hourly, daily, and monthly energy consumption. Analytical methods such as trend analysis, peak load detection, and energy intensity evaluation are applied to identify inefficient operating conditions. Threshold-based anomaly detection mechanisms are implemented to identify abnormal energy usage and potential equipment faults.

Historical data analysis enables comparison of energy performance across different machines, shifts, or operational periods. This analysis supports identification of recurring inefficiencies and optimization opportunities.

F. Dashboard Design and Visualization

An interactive industrial data dashboard is developed to present real-time and historical energy information in a user-friendly manner. Visualization components include line charts, bar graphs, pie charts, and key performance indicators (KPIs). The dashboard supports filtering and drill-down functionalities, allowing users to analyse energy consumption by time range, equipment, or production unit.

G. Alert Generation and Decision Support

Predefined energy thresholds are used to generate alerts when abnormal energy consumption or peak demand conditions are detected. Notifications assist operators in taking timely corrective actions to prevent energy wastage and equipment stress. The dashboard functions as a decision-support system by providing actionable insights that aid in energy optimization and operational planning.

H. System Evaluation and Scalability

The performance of the system is evaluated based on accuracy, responsiveness, and usability. The modular architecture enables easy scalability, allowing integration of additional sensors, analytical modules, or visualization components. The system is designed to adapt to varying industrial requirements and operational scales.

Workflow of AI-Integrated Industrial Data Dashboard & Predictive Maintenance System

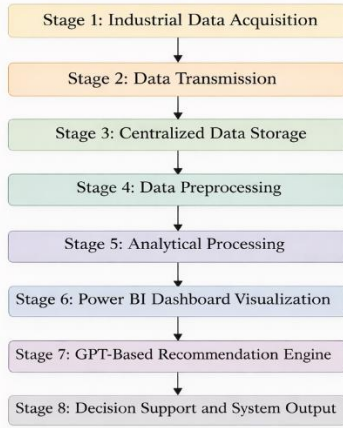


Fig.2 System Workflow Diagram

V. RESULT AND DISCUSSION

The proposed Industrial Data Dashboard and Predictive Maintenance System was validated using real-time and simulated industrial datasets acquired from energy meters and machine condition sensors. The system successfully captured electrical parameters including voltage, current, active power, power factor, and cumulative energy consumption, along with machine condition indicators such as temperature, vibration, and runtime hours. The data was transmitted to the centralized database with minimal latency, ensuring near real-time availability for analytical processing and visualization.

The Power BI dashboard effectively visualized time-series energy trends, peak load conditions, and comparative performance across monitored machines. During high operational intervals, a noticeable increase in active power and energy consumption was observed, confirming the system's capability to detect load variations and peak demand behaviour. The interactive filtering and drill-down features enabled detailed analysis of machine-specific performance, improving operational transparency.

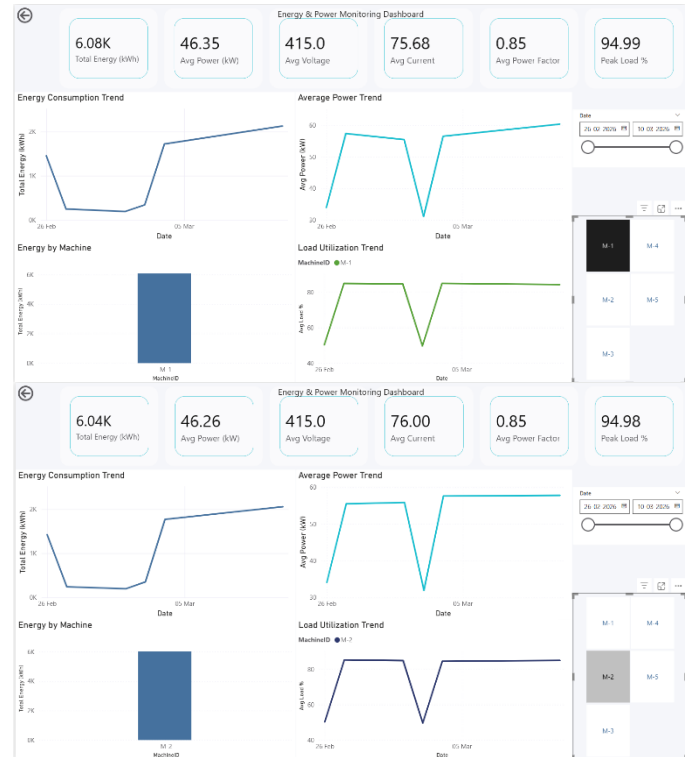
For predictive maintenance assessment, the degradation index derived from temperature, vibration, and runtime parameters successfully differentiated machine health conditions. Machines operating under elevated vibration and temperature levels exhibited higher degradation values and corresponding maintenance risk scores. The system generated automated alerts when threshold limits were exceeded, demonstrating effective anomaly detection and proactive maintenance planning capability.

A key enhancement of the proposed framework is the integration of the GPT-based intelligent recommendation engine. Instead of merely presenting numerical trends and risk indicators, the system analyzed summarized dashboard metrics and generated contextual recommendations. For example, low

power factor conditions triggered suggestions for corrective compensation, while abnormal vibration levels resulted in inspection advisories. During peak energy usage periods, the AI module recommended load balancing strategies to optimize energy consumption. This capability transformed the dashboard from a passive visualization tool into an intelligent advisory system.

Performance evaluation indicated stable communication, accurate visualization, and reliable analytical computation with system latency maintained within acceptable industrial limits. The combined integration of IoT-based monitoring, dashboard analytics, predictive modeling, and generative AI significantly enhanced decision-support functionality compared to conventional monitoring approaches.

Overall, the results validate that the proposed architecture provides a scalable, intelligent, and Industry 4.0-aligned solution for real-time energy management and predictive maintenance in industrial environments.



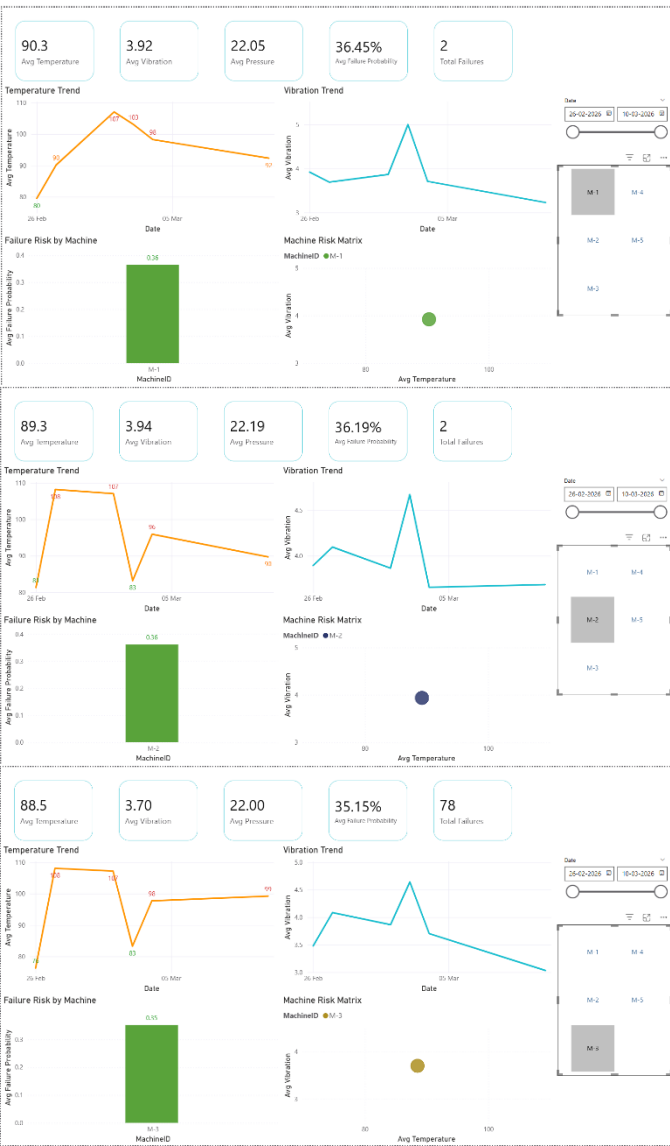
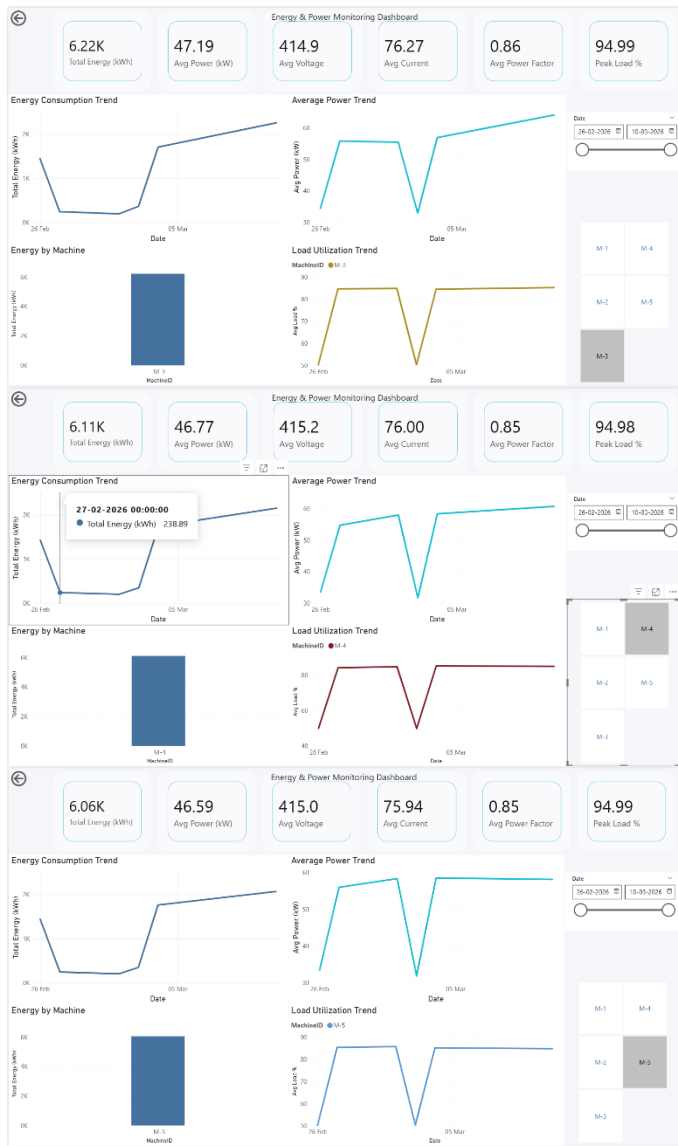


Fig. 1: shows the Energy and Power Monitoring Dashboard used to analyze the performance of multiple machines (M-1 to M-5). The dashboard displays key electrical parameters such as total energy consumption, average power, voltage, current, power factor, and peak load percentage. It also presents graphical trends including energy consumption trend, average power trend, energy distribution by machine, and load utilization trend over time. A date filter and machine selection panel allow users to analyze specific machines and time periods. This visualization helps operators monitor energy usage, identify abnormal patterns, and improve overall machine efficiency and energy management.



Fig. 2: The figure illustrates the industrial machine monitoring dashboard used for predictive maintenance analysis. It visualizes key operational parameters including average temperature, vibration, pressure, and failure probability for five machines (M1–M5). Time-series graphs present temperature and vibration trends over the selected date range, enabling the identification of abnormal patterns or sudden fluctuations in machine behavior. The failure risk chart highlights the probability of machine failure based on the analyzed parameters, while the machine risk matrix provides a comparative visualization of machine health conditions. This dashboard enables real-time monitoring and supports early fault detection, helping maintenance teams identify high-risk machines and schedule preventive maintenance to improve system reliability and operational efficiency.

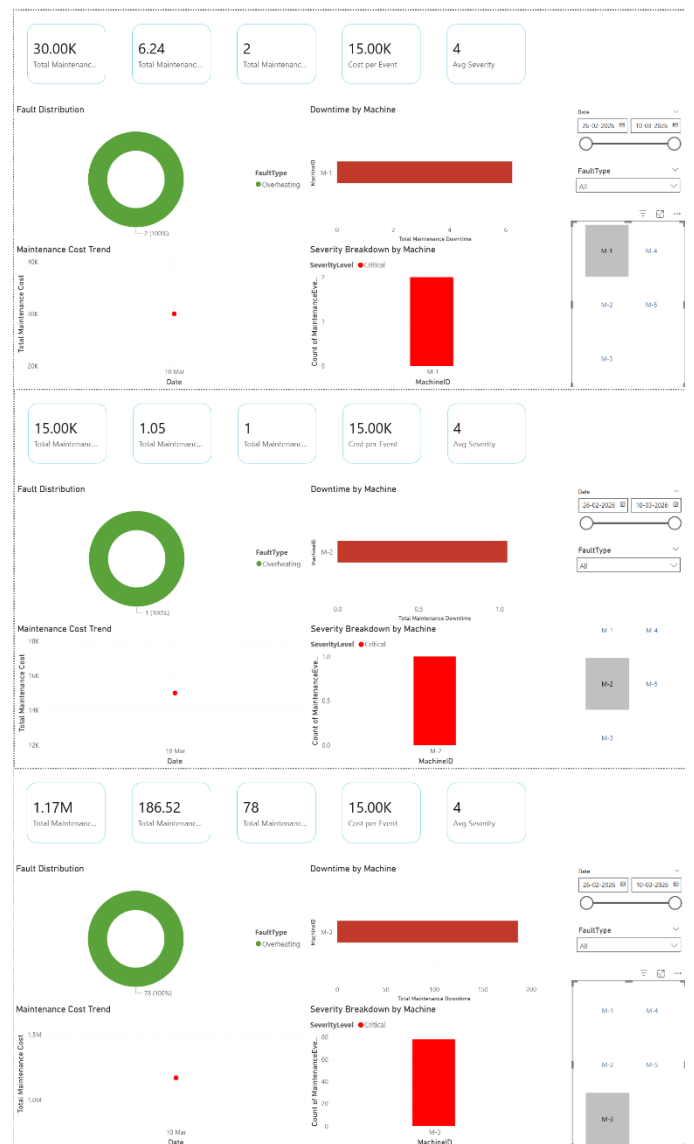




Fig. 3: Maintenance analytics dashboard for industrial machines.

The figure presents the maintenance analysis dashboard used to monitor fault conditions and maintenance performance for multiple machines (M1–M5). The dashboard displays key maintenance indicators such as total maintenance cost, total maintenance events, cost per event, and average severity level. Fault distribution charts highlight the type and frequency of machine faults, while downtime analysis shows the maintenance downtime associated with each machine. The maintenance cost trend provides insights into cost variations over time, and the severity breakdown illustrates the number of critical maintenance events for each machine. This visualization enables maintenance engineers to identify high-risk machines, analyze fault patterns, and optimize maintenance planning to reduce downtime and operational costs.





Fig. 4. Machine performance and maintenance cost analysis dashboard.

The figure presents the machine performance and maintenance analysis dashboard used for monitoring the operational condition of industrial machines (M1–M5). It displays key performance indicators such as total energy consumption, total maintenance cost, total downtime, average efficiency, and average failure probability for each machine. The machine performance comparison chart illustrates the relationship between energy usage and downtime across different machines, enabling comparative performance evaluation. The dashboard also includes maintenance cost trends and risk assessment indicators such as overall risk score, risk status, Mean Time Between Failures (MTBF), and Mean Time To Repair (MTTR). These visualizations help identify machines with higher downtime or maintenance cost, supporting data-driven decision-making for predictive maintenance and improved operational efficiency.

VI. CONCLUSION

This paper presented the design and implementation of an AI-integrated Industrial Data Dashboard and Predictive Maintenance System for smart industrial monitoring. The proposed framework combines real-time sensor-based data acquisition, centralized data storage, analytical processing, and interactive visualization using Power BI to enable effective monitoring of energy consumption and machine condition parameters.

The integration of a GPT-based recommendation engine enhances the system by transforming dashboard metrics into intelligent, context-aware suggestions for energy optimization and maintenance planning. Unlike conventional monitoring systems that only display numerical trends, the proposed solution provides automated interpretative insights, thereby improving operational decision-making efficiency.

Experimental validation confirms reliable data transmission, accurate visualization of energy trends, timely identification of abnormal machine conditions, and effective AI-generated recommendations. The modular and scalable architecture makes the system suitable for Industry 4.0 environments and adaptable to small- and medium-scale industrial applications.

Overall, the proposed system demonstrates that the integration of IoT, business intelligence dashboards, predictive analytics, and generative AI can significantly improve energy efficiency, reduce downtime, and support data-driven industrial management.

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